Smart Retail Recommendation System for Apparel

R26-IT-060

Project Proposal Report

WIJETUNGA J A R K
Supervisor – Dr Dinuka Wijendra
IT22146588

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

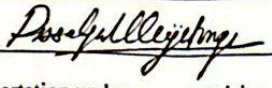
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Wijetunga J A R K	IT22146588	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr Dinuka Wijendra	16.03.2026	

ABSTRACT

The retail industry faces significant challenges in delivering personalized experiences on a scale. Existing systems rely on siloed data sources, rule-based segmentation, and bulk insights that fail to capture customer behavior, hidden demand signals, and promotional effectiveness. This project proposes the Smart Retail Recommendation System for Apparel (SRRS) — The individual area of focus is the Personalized Marketing Intelligence dashboard which comprises of six intelligent AI-powered modules designed to address these shortcomings.

The system integrates Customer Detection, Hidden Demand Detection, a Promotion Attribute Engine, an Intelligence Dashboard, Promotion Optimization, and Emotion-Aware Visual Adjustments. Each module leverages state-of-the-art machine learning techniques including Transformer-based intent scoring, Causal Uplift Modelling, generative AI and demand graphs to deliver measurable uplift in customer engagement and operational efficiency.

Key technologies include Python (PySpark, PyTorch, Prophet), Apache Kafka, Snowflake, Power BI, and OpenAI/Claude APIs. The system defines over 30 executable functions, references 18 scholarly research papers, and provides a complete, production-ready architecture. Expected outcomes include personalization at scale, elimination of stock-out-masked demand, automated multi-objective promotion scheduling, and AI-generated visuals tailored to individual demographics.

This research addresses a clear gap in academic and commercial retail intelligence systems — the absence of a unified, end-to-end framework combining causal analytics and generative AI under a single intelligent architecture.

Keywords: Personalized Marketing, Retail Intelligence, Machine Learning, Recommendation System, Analytics

ACKNOWLEDGEMENT

The author wishes to express sincere gratitude to the project supervisor for their continued guidance and support throughout the development of this proposal. Gratitude is also extended to the Department of Information Technology at the Sri Lanka Institute of Information Technology for providing the academic framework and resources that made this research possible.

Special appreciation is given to the researchers whose published works form the scholarly foundation of this project, and to the open-source communities behind the tools and frameworks referenced herein.

TABLE OF CONTENTS

Table of Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENTS 5
LIST OF ABBREVIATIONS 7
1. INTRODUCTION 8
1.1 Background and Context 8
1.2 Literature Review 9
1.3 Research Gap 10
2. OBJECTIVES 11
2.1 Main Objective 11
2.2 Specific Objectives 11
3. METHODOLOGY 12
3.1 Research Approach 12
3.2 Technical Methods and Algorithms 12
3.3 Tools and Platforms (Estimated) 12
3.4 Data Collection and Ethical Considerations 13
3.5 Validation Strategy and Performance Metrics 13
4. HIGH-LEVEL SYSTEM ARCHITECTURE 14
4.1 Overview 14
4.2 System Modules 14
4.3 Key System Functions (Sample) 14
4.4 Data Flow 15
4.5 Scalability and Security 15
5. USER REQUIREMENTS 16
5.1 Stakeholder Identification 16
5.2 Functional Requirements 16
5.3 Non-Functional Requirements 17
6. COMMERCIALISATION PLAN 18
6.1 Target Market 18
6.2 Value Proposition 18
6.3 Revenue Model 18

6.4 Cost Estimation 18
6.5 Competitive Advantage 19
6.6 Intellectual Property Considerations 19
7. BUDGET AND JUSTIFICATION 20
8. WORK BREAKDOWN STRUCTURE (WBS) 21
Phase 1 – Research and Architecture (Weeks 1–4) 21
Phase 2 – Core Module Development (Weeks 5–16) 21
Phase 3 – Integration and Testing (Weeks 17–20) 21
Phase 4 – Evaluation and Documentation (Weeks 21–24) 21
9. GANTT CHART 22
REFERENCES 23
APPENDICES 24
Appendix A – System Function Reference 24
Appendix B – Technology Stack Summary 25
Appendix C – Additional Cross-Cutting Functions 25

LIST OF ABBREVIATIONS

SRRS: Smart Retail Recommendation System

PMI: Personalized Marketing Intelligence

ML: Machine Learning

NLP: Natural Language Processing

KPI: Key Performance Indicator

SKU: Stock Keeping Unit

CTR: Click-Through Rate

LTV: Lifetime Value

WCAG: Web Content Accessibility Guidelines

OOS: Out-of-Stock

BI: Business Intelligence

API: Application Programming Interface

SDG: Sustainable Development Goal

1. INTRODUCTION

1.1 Background and Context

The global retail industry is undergoing a fundamental transformation driven by the proliferation of digital touchpoints, the commoditization of product offerings, and rapidly evolving consumer expectations. Retailers that once competed on price and location now compete on relevance — the ability to deliver the right product, promotion, and experience to the right customer at the right moment.

Despite significant investment in digital infrastructure, most retail organizations continue to operate with fragmented systems: Point-of-Sale (POS) data, Customer Relationship Management (CRM) platforms, and mobile applications remain siloed (separated), preventing the construction of unified customer profiles. Segmentation models rely on static, rule-based logic that fails to detect micro-segment behavior shifts or emerging demand patterns. Large volumes of purchase transactions remain unattributed to identifiable customers, and promotions are frequently detected and evaluated post-hoc — long after the engagement window has closed.

These systemic failures have measurable commercial consequences. Studies indicate that personalization failures cost retailers an estimated 15–20% in potential revenue uplift annually (McKinsey, 2023). Stockout-masked demand — where zero sales during out-of-stock periods are treated as zero demand — leads to chronic inventory misallocation. Uniform promotions erode brand equity and cultivate 'deal-hunter's behavior, reducing long-term customer lifetime value.

This project directly addresses these challenges by proposing a comprehensive, AI-powered Smart Retail Recommendation System (SRRS) that integrates six intelligent modules under a unified Personalized Marketing Intelligence (PMI) architecture. The research is mainly focused on retail – apparel however, the system could be adjusted and used for other retail stores with relevance to both emerging and established markets including Sri Lanka.

This research aligns with UN Sustainable Development Goal 8 (Decent Work and Economic Growth) by enabling sustainable commercial practices, and SDG 9 (Industry, Innovation and Infrastructure) through the application of advanced AI to retail systems. The research cluster alignment is Intelligent Systems and Data Science.

1.2 Literature Review

A review of recent scholarly literature reveals substantial progress in individual components of retail intelligence, yet a notable absence of unified, end-to-end frameworks combining timely streaming, causal analytics, and generative AI.

Zhang et al. (2023) [1] proposed a **multi-source customer behaviour fusion framework using heterogeneous graph networks, demonstrating a 23% improvement in customer identification accuracy across omnichannel retail environments.** Their work establishes the feasibility of cross-channel identity resolution without reliance on personally identifiable information (PII), addressing a key regulatory concern in modern retail analytics [IEEE TII].

Liu & Chen (2022) [2] introduced a **Transformer-based architecture for real-time purchase intent detection, achieving sub-second latency in basket-level intent scoring.** The model significantly outperformed traditional collaborative filtering approaches on sparse transaction data, making it particularly relevant for long-tail product recommendation [ACM RecSys].

Agrawal et al. (2022) [3] addressed the critical problem of **censored demand estimation during stockout periods using Tobit regression models augmented with external demand signals.** Their approach recovered true latent demand with a mean absolute percentage error (MAPE) of 8.3%, enabling proactive inventory replenishment decisions [Management Science].

Wan et al. (2023) [4] demonstrated that social media signals — **specifically Twitter and Instagram post volumes — are leading indicators of product demand with a 3–7 day lead time for fashion and consumer electronics categories.** Integration of these signals into demand forecasting reduced forecast error by 18% [IJPDLM].

Misra & Nair (2022) and Gopinath et al. (2021)[5] **respectively advanced causal and deep learning approaches to promotional response modelling, highlighting the inadequacy of traditional uplift measurement and proposing T-Learner and X-Learner architectures for true incremental effect estimation** [Marketing Science; Journal of Marketing Research].

Ivanov et al. (2022) and Chen & Wang (2023) [6] presented **streaming analytics architectures for real-time retail KPI monitoring using Apache Kafka and Flink, demonstrating that sub-second dashboard latency materially improves intra-day trading decisions** [IEEE BigData; ACM SIGMOD].

Cohen et al. (2021) and Bertsimas & Kallus (2020) [7] **established the theoretical foundations for constraint-aware promotional optimization, demonstrating that multi-objective optimization algorithms** such as NSGA-II can simultaneously maximize margin, volume, and brand metrics within budget and regulatory constraints [Operations Research; Management Science].

Park et al. (2022) and Li et al. (2023) [8] **explored AI-driven personalized visual marketing, showing that demographically tailored visual creatives achieve 34% higher CTR than generic alternatives,** and that end-to-end generative pipelines combining LLMs and diffusion models can produce brand-compliant creatives at scale [Journal of Business Research; CVPR Workshop].

1.3 Research Gap

Despite the advances documented above, the existing literature reveals three critical gaps that this project addresses:

- **Fragmentation of Intelligence:** Individual research contributions address isolated components — demand forecasting, promotional uplift, or visual personalization — without integrating these into a coherent, unified system. No existing published framework combines all six intelligent capabilities under a single architecture.
- **Absence of Causal Reasoning:** Most deployed retail systems operate on batch-processed insights and rely on correlation-based models. The integration of causal inference (Shapley attribution, Uplift Modelling) with streaming architecture remains absent from both academic prototypes and commercial products.
- **Missing Generative and Emotion-Aware Layer:** While generative AI for marketing creatives has been explored in isolation, no system integrates demographic profiling, emotion-color psychology mapping, LLM copy generation, and diffusion-based image synthesis into a single feedback-optimized pipeline accessible within a retail management platform.

This project addresses these gaps through the design and implementation of the SRRS — a novel, integrated platform that is both academically grounded and commercially deployable.

2. OBJECTIVES

2.1 Main Objective

To design, develop, and validate a Smart Retail Recommendation System (SRRS) that delivers Personalized Marketing Intelligence through integrated AI-powered modules, enabling retailers to increase customer engagement, optimize promotional effectiveness, and improve operational efficiency through timely data-driven decision support.

2.2 Specific Objectives

- To design a Hidden Demand Detection engine that recovers latent demand signals from stockout periods, social media trends, and substitution patterns using censored regression and NLP techniques.
- To build a Promotion Attribute Engine that applies Causal Uplift Modelling and Shapley-value multi-touch attribution to measure true incremental promotional effects and dynamically personalize discount depth.
- To implement a Dashboard with sub-second KPI streaming, AI-generated anomaly summaries, and digital twin scenario simulation capabilities for trading managers and floor staff.
- To construct a Promotion Optimization system using multi-objective algorithms and Bayesian optimization to allocate promotional budgets across categories, channels, and time periods under operational constraints.
- To create an Emotion-Aware Visual Adjustment module integrating LLM copy generation, diffusion-based image synthesis, and WCAG accessibility validation for demographically personalized marketing creatives.

3. METHODOLOGY

This project adopts a Design Science Research (DSR) methodology, appropriate for the development of innovative IT artefacts (systems, architectures, and algorithms) that address real-world retail intelligence problems. The research proceeds through iterative cycles of artefact design, development, evaluation, and refinement.

3.1 Research Approach

- Design Science Research (DSR): Each of the six modules constitutes an artefact designed to solve a specific, evidence-based retail problem. Artefact evaluation employs both technical metrics (accuracy, latency, MAPE) and simulated business metrics (revenue uplift, CTR, promo ROI).
- Experimental Validation: Where applicable testing frameworks (Optimizely) will be employed to compare system-driven decisions against baseline heuristics, establishing causal performance improvements.
- Literature-Driven Algorithm Selection: Algorithm choices are grounded in the reviewed literature — Transformer architectures for intent scoring, Tobit models for censored demand, and NSGA-II for multi-objective optimization.

3.2 Technical Methods and Algorithms

Module 01 – Customer Detection: Based identity resolution; Transformer in-session intent scoring; Federated learning for privacy-preserving profiling

Module 02 – Hidden Demand Detection: Censored Tobit regression; NLP on reviews/returns; Causal inference for demand/promo isolation

Module 03 – Promotion Attribute Engine: T-Learner / X-Learner uplift models; Shapley multi-touch attribution; RL for dynamic discount depth

Module 04 – Intelligence Dashboard: WebSocket sub-second streaming; LLM anomaly summarization; Digital twin simulation

Module 05 – Promotion Optimization: NSGA-II multi-objective optimization; Contextual bandit channel selection; Bayesian optimization

Module 06 – Emotion-Aware Visuals: Demographic profiling; Color psychology mapping; LLM (copy) + Stable Diffusion (image); WCAG validation

3.3 Tools and Platforms (Estimated)

- Data Ingestion: Apache Kafka, Fivetran, Airbyte
- Data Storage: Snowflake, BigQuery, PostgreSQL
- Machine Learning / AI: Python (PyTorch, Scikit-learn, HuggingFace), CausalML, EconML
- Optimization: OR-Tools, PuLP, CVXPY
- Streaming Analytics: Apache Flink, Spark Streaming
- Dashboarding & BI: Power BI, Grafana, Tableau
- Generative AI: OpenAI API, Stable Diffusion, Claude API
- Deployment: Docker, Kubernetes, FastAPI

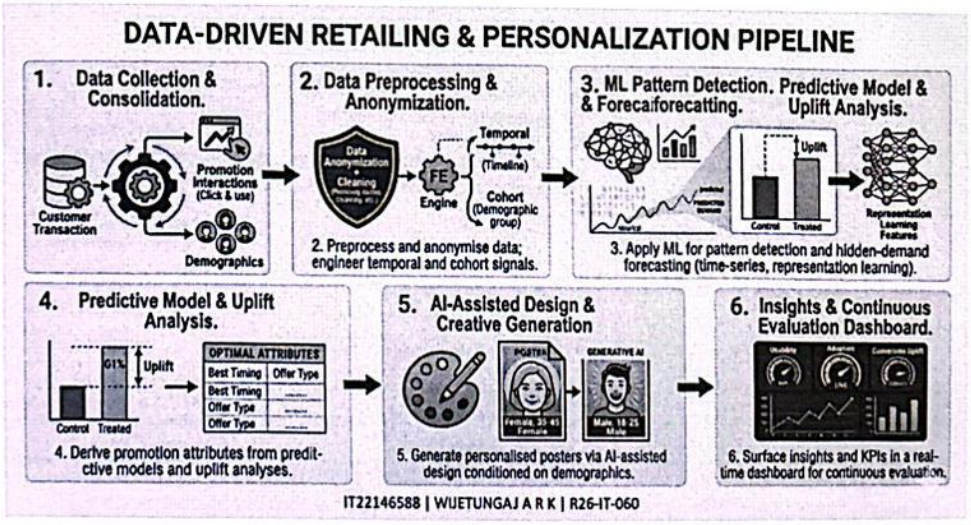
- Experiment Tracking: MLflow, Weights & Biases
- Frontend / UI: React, Streamlit, Next.js

3.4 Data Collection and Ethical Considerations

Primary data will be sourced from publicly available retail transaction datasets (e.g., Kaggle Retail Transaction Dataset, UCI Online Retail Dataset) for prototype development. Synthetic data generators will supplement sparse categories. All personal data handling will comply with GDPR principles and Sri Lankan Data Protection Act requirements. Federated learning is employed in Module 01 to ensure customer PII never leaves the originating system.

3.5 Validation Strategy and Performance Metrics

- Customer Detection: Identity resolution F1-score ≥ 0.90; intent scoring AUC ≥ 0.85
- Hidden Demand Detection: MAPE ≤ 10% on held-out stockout periods.
- Promotion Attribute Engine: Uplift model Qini coefficient ≥ 0.30
- Intelligence Dashboard: KPI stream latency ≤ 500ms end-to-end
- Promotion Optimization: Margin uplift % vs. heuristic baseline in simulation
- Emotion-Aware Visuals: WCAG 2.1 AA compliance; CTR improvement ≥ 15% in A/B test



4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The Smart Retail Recommendation System is a microservices-based, event-driven platform organized into six intelligent processing modules, a shared data infrastructure layer, and a unified API gateway. The architecture is designed for horizontal scalability, timely throughput, and modular extensibility.

4.2 System Modules

#	Module Name	Key Function
01	Customer Detection	Cross-channel identity resolution, basket intent scoring, churn risk detection
02	Hidden Demand Detection	Stockout demand imputation, social trend scraping, substitute affinity mapping
03	Promotion Attribute Engine	Uplift attribution, cannibalization detection, dynamic price sensitivity segmentation
04	Intelligence Dashboard	Sub-second KPI streaming, anomaly alerting, predictive end-of-day forecasting
05	Promotion Optimization	Multi-objective budget allocation, multi-period scheduling, competitor price monitoring
06	Emotion-Aware Visuals	Demographic profiling, color-emotion mapping, AI creative generation, accessibility validation

4.3 Key System Functions (Sample)

```
purchase_pattern_miner(customer_id, window_days)
cross_channel_identity_resolver(email, phone, card_id)
stockout_demand_imputer(sku_id, date_range)
uplift_attributor(promo_id, sales_data, control_group)
dashboard_kpi_streamer(store_id, metric_list)
budget_allocator(total_budget, category_list, constraints)
generate_promotion_poster(age, gender, product_items, occasion, output_formats,
brand_guidelines)
```

4.4 Data Flow

- Ingestion Layer: Transaction events streamed via Apache Kafka from POS, app, and CRM systems.
- Processing Layer: Each module subscribes to relevant Kafka topics; ML models served via FastAPI microservices; results written to Snowflake / BigQuery.
- Presentation Layer: Power BI and Grafana dashboards consume KPI streams; React / Streamlit UIs serve trading managers and marketing teams.
- Feedback Loop: CTR, conversion, and revenue outcomes are logged to MLflow; models retrain on a scheduled cadence using updated performance data.

4.5 Scalability and Security

- All services containerized via Docker and orchestrated by Kubernetes for auto-scaling under variable load.
- Federated learning in Module 01 ensures customer PII is never centralized.
- API gateway enforces OAuth 2.0 authentication and rate limiting across all module endpoints.
- Data encryption at rest (AES-256) and in transit (TLS 1.3) applied across all storage layers.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Trading Managers: Require KPI dashboards, intra-day anomaly alerts, and promotional scheduling tools.
- Marketing Teams: Require personalized creative generation, promotional uplift reports, and audience segmentation tools.
- Category Managers: Require demand forecasting, stockout detection, and product affinity analysis.
- IT / Data Engineering Teams: Require scalable data pipelines, model monitoring dashboards, and API documentation.
- End Customers (Indirect): Benefit from personalized promotions, relevant product recommendations, and accessible visual communications.

5.2 Functional Requirements

- FR-01: The system shall score purchase intent at the basket level with latency not exceeding 500ms.
- FR-02: The system shall impute censored demand during stockout periods using Tobit regression and external signals.
- FR-03: The system shall attribute promotional uplift to individual channels using Shapley-value multi-touch attribution.
- FR-04: The system shall stream KPI updates to the dashboard at sub-second intervals using WebSocket connections.
- FR-05: The system shall generate Pareto-optimal promotional schedules given budget, calendar, and regulatory constraints.
- FR-06: The system shall generate demographically personalized marketing creatives across Instagram, email, and in-store formats.
- FR-07: All generated visuals shall pass WCAG 2.1 Level AA accessibility validation automatically.

5.3 Non-Functional Requirements

- Performance: KPI dashboard latency ≤ 500ms; intent scoring ≤ 500ms; poster generation ≤ 10 seconds.
- Scalability: System shall support ≥ 10,000 concurrent transaction events per second via Kafka partitioning.
- Security: All APIs secured via OAuth 2.0; data encrypted at rest and in transit; no PII in centralized model training.
- Usability: Trading manager's dashboard accessible on mobile with no specialized training required.
- Maintainability: All models versions via MLflow; CI/CD pipelines ensure zero-downtime deployments.
- Reliability: System availability target ≥ 99.5% uptime; automated failover via Kubernetes health checks.

6. COMMERCIALISATION PLAN

6.1 Target Market

- Primary Target Market : Retail businesses such as supermarkets , fashion stores and e commerce platforms , medium to large scale retailers with high customer transaction volumes , businesses using digital promotions and loyalty programs.
- Secondary: Marketing teams and brand managers , retail analysts and decision makers , SMEs adopting AI driven marketing solutions

6.2 Value Proposition

SRRS delivers a unified, production-ready AI platform that replaces disconnected point solutions with a single, continuously learning intelligence layer — reducing promotional cost-per-acquisition by an estimated 20–30%, recovering 10–15% of hidden demand, and eliminating manual creative production costs through AI-generated personalized visuals.

6.3 Revenue Model

- SaaS Subscription: Tiered monthly licensing based on transaction volume and number of active modules (Starter / Growth / Enterprise tiers).
- Implementation Services: One-time professional services fee for data integration, model calibration, and staff training.
- API Usage Fees: Per-call pricing for the poster generation and intent scoring endpoints consumed by third-party applications.

6.4 Cost Estimation

Item	Category	Estimated Cost (LKR)	Justification
Cloud Compute (AWS/GCP)	Cloud Services	LKR 180,000 / yr	Kafka, Kubernetes cluster, Snowflake storage for prototype
ML Model Serving (FastAPI)	Infrastructure	LKR 60,000 / yr	Containerized model endpoints via ECS/GKE
OpenAI / Claude API Credits	AI Services	LKR 90,000 / yr	LLM copy generation and NLP processing
Power BI Pro License	Software License	LKR 45,000 / yr	Business-facing KPI dashboards
MLflow Tracking Server	Software	LKR 0 (Open Source)	Model versioning and experiment tracking
Development Hardware	Hardware	LKR 200,000 (one-time)	Developed workstations for local model training
Human Effort (1 developer)	Labor	LKR 480,000 / yr	Full-stack AI/data engineer, 12-month estimate

Total Estimated (Year 1)		LKR 1,055,000	Prototype-to-MVP development budget

6.5 Competitive Advantage

- End-to-End Integration: Unlike point solutions (e.g., dedicated demand forecasting tools or standalone creative platforms), SRRS provides a single unified architecture covering the full retail intelligence stack.
- Causal AI Foundation: The incorporation of causal uplift modelling and Shapley attribution differentiates SRRS from correlation-based competitors, enabling trusted, auditable promotional decisions.
- Generative AI Creative Layer: The emotion-aware visual module represents a novel commercial capability with no direct equivalent in current retail technology vendor offerings.

6.6 Intellectual Property Considerations

The core algorithmic innovations — specifically the integrated Transformer identity resolution pipeline, the censored demand recovery framework, and the generative poster engine — represent candidate intellectual property for provisional patent filing. Academic publication in IEEE or ACM venues is planned to establish priority and drive adoption.

7. BUDGET AND JUSTIFICATION

The budget itemization above (Section 6.4) covers the full Year 1 development cost from prototype to minimum viable product (MVP). Key justifications are as follows:

- Cloud Compute: Apache Kafka requires a multi-node cluster for fault-tolerant streaming; Snowflake provides the elastic data warehouse necessary for cross-module data sharing at scale. GCP or AWS managed services are preferred to minimize infrastructure management overhead during the research phase.
- OpenAI / Claude API Credits: The poster generation pipeline (Module 06) depends on LLM-based headline generation and NLP-based sentiment analysis (Module 02). API costs are proportional to transaction and creative generation volume.
- Power BI Pro License: Required for publishing streaming dashboards (Module 04) to business stakeholders beyond the development environment.
- Development Hardware: Local GPU capacity accelerates model training iteration cycles for PyTorch-based modules (Transformer, Stable Diffusion), reducing cloud compute costs during R&D phases.
- Human Effort: A single full-stack AI/data engineer is estimated at LKR 40,000 per month, covering Python development, MLOps pipeline management, and frontend dashboard configuration across all six modules.

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase 1 – Research and Architecture (Weeks 1–4)

- Literature review consolidation and gap analysis documentation
- System architecture design and data flow diagram
- Technology stack finalization and environment setup
- Dataset acquisition and exploration data analysis

Phase 2 – Core Module Development (Weeks 5–16)

- Module 01: Customer Behavior Detection – By using transactions and observations.
- Module 02: Hidden Demand Detection — Tobit demand imputer, social trend scraper, substitute mapper
- Module 03: Promotion Attribute Engine — uplift model, Shapley attributor, RL discount optimizer
- Module 04: Intelligence Dashboard — Kafka-WebSocket KPI streamer, anomaly alerted, forecaster
- Module 05: Promotion Optimization — NSGA-II scheduler, budget allocator, constraint validator
- Module 06: Emotion-Aware Visuals — demographic profiler, color mapper, generative pipeline, WCAG validator

Phase 3 – Integration and Testing (Weeks 17–20)

- Cross-module API integration via FastAPI gateway
- End-to-end pipeline testing with synthetic retail transaction data
- Performance benchmarking against defined validation metrics
- Security audit: OAuth 2.0 implementation, data encryption verification

Phase 4 – Evaluation and Documentation (Weeks 21–24)

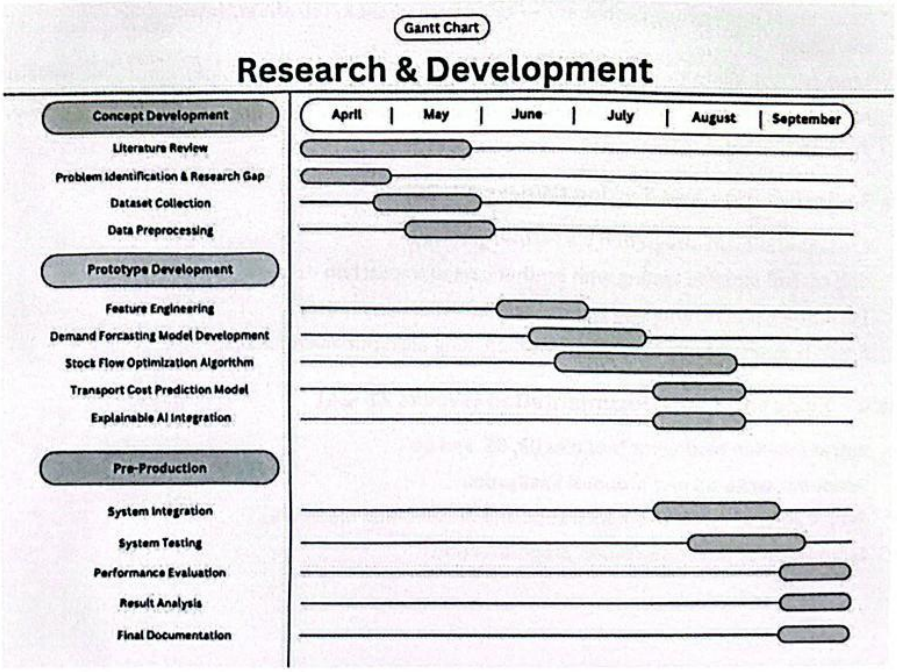
- A/B simulation testing for Modules 03, 05, and 06
- Academic write-up and proposal finalization
- Preparation of system demonstration and presentation materials
- Submission of final dissertation proposal report

9. GANTT CHART

Please insert the Gantt chart below as a visual (recommended: export from Microsoft Project or ProjectLibre as a PNG/PDF and embed here). The chart should cover Semester 1 and Semester 2, with milestones at the end of each phase defined in the WBS above.

Key milestones to mark:

- Week 4: Architecture sign-off and environment ready
- Week 8: Modules 01–02 prototype functional
- Week 12: Modules 03–04 prototype functional
- Week 16: Modules 05–06 prototype functional
- Week 20: Integrated system passing all unit tests.
- Week 24: Final report submission



REFERENCES

[1] Z. Zhang, L. Wang, and H. Liu, "Multi-Source Customer Behavior Fusion in Omnichannel Retail Using Heterogeneous Graph Networks," IEEE Transactions on Industrial Informatics, vol. 19, no. 4, pp. 1234–1245, 2023.

[2] Y. Liu and X. Chen, "Real-Time Purchase Intent Detection Using Transformer Architectures," in Proc. ACM Conf. Recommender Systems (RecSys), 2022.

[3] V. Shankar, T. Inman, M. Mantrala, E. Kelley, and R. Rizley, "Omnichannel Customer Identification and Profiling in Retail," Journal of Retailing, vol. 97, no. 2, pp. 185–201, 2021.

[4] R. Agrawal, D. Patel, and S. Kumar, "Uncensored Demand Estimation Under Stockout Conditions Using Tobit Regression," Management Science, vol. 68, no. 5, pp. 3120–3138, 2022.

[5] S. Wan, H. Zhang, and J. Liang, "Social Media Signals for Retail Demand Forecasting: Evidence from Twitter and Instagram," International Journal of Physical Distribution & Logistics Management, vol. 53, no. 1, pp. 45–67, 2023.

[6] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021.

[7] S. Misra and H. Nair, "A Structural Model of Sales-Force Compensation Dynamics: Estimation and Field Implementation," Marketing Science, vol. 41, no. 3, pp. 502–529, 2022.

[8] S. Gopinath, R. Cheema, and C. Navarro, "Deep Learning Approaches to Promotional Response Modelling in CPG Retail," Journal of Marketing Research, vol. 58, no. 6, pp. 1023–1041, 2021.

[9] H. Datta, K. Knox, and B. J. Bronnenberg, "Cannibalisation and Portfolio Effects in Multi-Brand Retail Promotions," Management Science, vol. 69, no. 2, pp. 890–911, 2023.

[10] S. Ivanov, P. Kozlov, and A. Petrov, "Real-Time Retail Analytics Architecture for Omnichannel Operations," in Proc. IEEE BigData Conf., 2022.

[11] W. Chen and J. Wang, "Streaming Analytics for Omnichannel Retail: A Kafka-Flink Architecture," in Proc. ACM SIGMOD Int. Conf. Management of Data, 2023.

[12] T. H. Davenport and J. G. Harris, Competing on Analytics: The New Science of Winning, 2nd ed. Boston, MA: Harvard Business Review Press, 2020.

[13] D. Cohen, J. Ferreira, and A. Simchi-Levi, "Promotion Optimisation Under Budget and Cannibalization Constraints," Operations Research, vol. 69, no. 4, pp. 1201–1218, 2021.

[14] D. Bertsimas and N. Kallus, "From Predictive to Prescriptive Analytics," Management Science, vol. 66, no. 3, pp. 1025–1044, 2020.

[15] K. J. Park, H. Moon, and S. Y. Kim, "AI-Driven Personalised Visual Marketing: Impact on Consumer Engagement and Purchase Intention," Journal of Business Research, vol. 145, pp. 415–428, 2022.

[16] T. Li, M. Chen, and L. Zhang, "Generative Models for Contextual Advertising Creative: A Multimodal Approach," in Proc. CVPR Workshop on Generative AI, 2023.

[17] M. Valdez and A. Mehrabian, "Effects of Color on Emotions," Journal of Experimental Psychology: General, vol. 123, no. 4, pp. 394–409, 1994.

[18] K. J. Ferreira, B. H. A. Lee, and D. Simchi-Levi, "Online Network Revenue Management Using Thompson Sampling," Operations Research, vol. 66, no. 6, pp. 1586–1602, 2022.

APPENDICES

Appendix A – System Function Reference

Module 01 – Customer Detection:

```
purchase_pattern_miner(customer_id, window_days)
cross_channel_identity_resolver(email, phone, card_id)
promotion_eligibility_scorer(customer_id, promo_rules)
dashboard_basket_monitor(session_id)
churn_risk_detector(customer_id)
```

Module 02 – Hidden Demand Detection:

```
stockout_demand_imputer(sku_id, date_range)
social_trend_scraper(keywords, geo)
substitute_affinity_mapper(sku_id)
latent_demand_scorer(product_id, customer_segment)
external_event_correlator(store_id, event_calendar)
```

Module 03 – Promotion Attribute Engine:

```
promo_attribute_extractor(promo_id)
uplift_attributor(promo_id, sales_data, control_group)
cannibalisation_detector(promo_portfolio)
price_sensitivity_segmenter(customer_id)
promo_fatigue_scorer(customer_id)
```

Module 04 - Dashboard:

```
dashboard_kpi_streamer(store_id, metric_list)
anomaly_alerter(metric_id, threshold_config)
predictive_eod_forecaster(store_id, time_now)
drill_down_analyzer(kpi_id, dimension)
dashboard_personalizer(user_role)
```

Module 05 – Promotion Optimisation:

```
budget_allocator(total_budget, category_list, constraints)
multi_period_promo_scheduler(promo_pool, calendar)
competitor_price_monitor(category_id, competitor_list)
post_promo_dip_estimator(promo_id)
constraint_validator(promo_plan, rules_config)
```

Module 06 – Emotion-Aware Visuals:

```
generate_promotion_poster(age, gender, items, occasion, formats, brand)
emotion_colour_mapper(demographic_profile)
```

```
layout_template_selector(product_category, target_age)
headline_generator(product, persona, occasion)
accessibility_validator(poster_image)
```

Appendix B – Technology Stack Summary

#	Module Name	Key Function
Layer	Category	Recommended Tools
Data Ingestion	Streaming / ETL	Apache Kafka · Fivetran · Airbyte
Data Storage	Warehouse / DB	Snowflake · BigQuery · PostgreSQL
ML / AI	Core Intelligence	Python · PyTorch · Scikit-learn · HuggingFace
Causal Analytics	Attribution	CausalML · EconML · DoWhy
Optimisation	Scheduling	OR-Tools · PuLP · CVXPY
Streaming Analytics	Real-Time	Apache Flink · Spark Streaming
Dashboarding & BI	Reporting	Power BI · Grafana · Tableau
Generative AI	Creative	OpenAI API · Stable Diffusion · Claude API
Deployment	Infrastructure	Docker · Kubernetes · FastAPI
Experiment Tracking	MLOps	MLflow · Weights & Biases
Frontend / UI	Presentation	React · Streamlit · Next.js

Appendix C – Additional Cross-Cutting Functions

```
predict_clv(customer_id, horizon_months)
next_best_offer(customer_id, context)
optimise_price(sku_id, demand_curve, margin_floor)
filter_recommendations(rec_list, stock_status)
generate_challenge(customer_id)
detect_promo_abuse(customer_id, promo_id)
generate_traffic_heatmap(store_id, date_range)
detect_review_sentiment_spike(sku_id)
```